

Web Analytics & Digital Marketing Analysis (PostgreSQL)

This document contains a complete SQL-based Web Analytics and Digital Marketing analysis project structured in 5 phases: Traffic Analysis, Conversion Analysis, Funnel Analysis, Product & Profitability Analysis, and Strategic Insights.

PHASE 1 – Traffic & Session Analysis

1. Monthly Sessions Trend

```
SELECT
    DATE_TRUNC('month', created_at) AS month,
    COUNT(*) AS total_sessions
FROM website_sessions
GROUP BY 1
ORDER BY 1;
```

2. Sessions by Traffic Source

```
SELECT
    utm_source,
    utm_campaign,
    COUNT(*) AS sessions
FROM website_sessions
GROUP BY 1,2
ORDER BY sessions DESC;
```

3. Device Breakdown

```
SELECT
    device_type,
    COUNT(*) AS sessions,
    ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER(), 2) AS pct_of_total
FROM website_sessions
GROUP BY device_type;
```

4. New vs Repeat Sessions

```
SELECT
    is_repeat_session,
    COUNT(*) AS sessions,
    ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER(), 2) AS pct_share
FROM website_sessions
GROUP BY is_repeat_session;
```

PHASE 2 – Conversion Analysis

5. Overall Conversion Rate

```
SELECT
    COUNT(DISTINCT ws.website_session_id) AS sessions,
    COUNT(DISTINCT o.order_id) AS orders,
    ROUND(
        COUNT(DISTINCT o.order_id)::decimal
        / COUNT(DISTINCT ws.website_session_id) * 100,
        2) AS conversion_rate_pct
FROM website_sessions ws
LEFT JOIN orders o
    ON ws.website_session_id = o.website_session_id;
```

6. Conversion Rate by Traffic Source

```
SELECT
    ws.utm_source,
    ws.utm_campaign,
    COUNT(DISTINCT ws.website_session_id) AS sessions,
    COUNT(DISTINCT o.order_id) AS orders,
    ROUND(
        COUNT(DISTINCT o.order_id)::decimal
        / COUNT(DISTINCT ws.website_session_id) * 100,
        2) AS conversion_rate_pct
FROM website_sessions ws
LEFT JOIN orders o
    ON ws.website_session_id = o.website_session_id
GROUP BY 1,2
ORDER BY conversion_rate_pct DESC;
```

7. Conversion Rate by Device

```
SELECT
  ws.device_type,
  COUNT(DISTINCT ws.website_session_id) AS sessions,
  COUNT(DISTINCT o.order_id) AS orders,
  ROUND(
    COUNT(DISTINCT o.order_id)::decimal
    / COUNT(DISTINCT ws.website_session_id) * 100,
    2) AS conversion_rate_pct
FROM website_sessions ws
LEFT JOIN orders o
  ON ws.website_session_id = o.website_session_id
GROUP BY 1;
```

8. Revenue & AOV

```
SELECT
  COUNT(DISTINCT o.order_id) AS total_orders,
  SUM(o.price_usd) AS total_revenue,
  ROUND(AVG(o.price_usd),2) AS avg_order_value
FROM orders o;
```

9. Revenue per Session

```
SELECT
  ROUND(
    SUM(o.price_usd)::decimal
    / COUNT(DISTINCT ws.website_session_id),
    2) AS revenue_per_session
FROM website_sessions ws
LEFT JOIN orders o
  ON ws.website_session_id = o.website_session_id;
```

PHASE 3 – Funnel Analysis

10. Clickstream Analysis

```
SELECT
  website_session_id,
  COUNT(*) AS total_pageviews,
  STRING_AGG(DISTINCT pageview_url, ' -> ' ORDER BY pageview_url) AS unique_pages
```

```
FROM
    website_page_views
GROUP BY
    website_session_id
ORDER BY
    website_session_id;
```

11. Funnel Step Flags

```
WITH session_page_flags AS (
    SELECT
        website_session_id,
        MAX(CASE WHEN pageview_url = '/home' THEN 1 ELSE 0 END) AS home_page,
        MAX(CASE WHEN pageview_url = '/products' THEN 1 ELSE 0 END) AS products_page,
        MAX(CASE WHEN pageview_url = '/cart' THEN 1 ELSE 0 END) AS cart_page,
        MAX(CASE WHEN pageview_url = '/shipping' THEN 1 ELSE 0 END) AS shipping_page,
        MAX(CASE WHEN pageview_url = '/billing' THEN 1 ELSE 0 END) AS billing_page
    FROM website_page_views
    GROUP BY website_session_id
)
SELECT * FROM session_page_flags;
```

12. Funnel Drop-Off Percentage

```
WITH session_page_flags AS (
    SELECT
        website_session_id,
        MAX(CASE WHEN pageview_url = '/home' THEN 1 ELSE 0 END) AS home_page,
        MAX(CASE WHEN pageview_url = '/products' THEN 1 ELSE 0 END) AS products_page,
        MAX(CASE WHEN pageview_url = '/cart' THEN 1 ELSE 0 END) AS cart_page,
        MAX(CASE WHEN pageview_url = '/shipping' THEN 1 ELSE 0 END) AS shipping_page,
        MAX(CASE WHEN pageview_url = '/billing' THEN 1 ELSE 0 END) AS billing_page
    FROM website_page_views
    GROUP BY website_session_id
)
SELECT
    ROUND(100.0 * SUM(products_page) / SUM(home_page), 2) AS home_to_products_pct,
    ROUND(100.0 * SUM(cart_page) / SUM(products_page), 2) AS products_to_cart_pct,
    ROUND(100.0 * SUM(shipping_page) / SUM(cart_page), 2) AS cart_to_shipping_pct,
    ROUND(100.0 * SUM(billing_page) / SUM(shipping_page), 2) AS shipping_to_billing_pct
FROM session_page_flags;
```

PHASE 4 – Product & Profitability Analysis

13. Revenue by Product

```
SELECT
    p.product_name,
    SUM(oi.price_usd) AS revenue,
    SUM(oi.cogs_usd) AS total_cost,
    SUM(oi.price_usd - oi.cogs_usd) AS gross_profit
FROM order_items oi
JOIN products p
    ON oi.product_id = p.product_id
GROUP BY 1
ORDER BY revenue DESC;
```

14. Refund Rate by Product

```
SELECT
    p.product_name,
    COUNT(DISTINCT oi.order_item_id) AS items_sold,
    COUNT(DISTINCT r.order_item_id) AS items_refunded,
    ROUND(
        100.0 * COUNT(DISTINCT r.order_item_id)
        / COUNT(DISTINCT oi.order_item_id),
        2) AS refund_rate_pct
FROM order_items oi
LEFT JOIN order_item_refunds r
    ON oi.order_item_id = r.order_item_id
JOIN products p
    ON oi.product_id = p.product_id
GROUP BY 1
ORDER BY refund_rate_pct DESC;
```

15. Net Revenue After Refunds

```
SELECT
    SUM(o.price_usd) AS gross_revenue,
    COALESCE(SUM(r.refund_amount_usd),0) AS total_refunds,
    SUM(o.price_usd) - COALESCE(SUM(r.refund_amount_usd),0) AS net_revenue
FROM orders o
LEFT JOIN order_item_refunds r
    ON o.order_id = r.order_id;
```

PHASE 5 – Strategic Insights

16. Conversion Rate: New vs Repeat Users

```
SELECT
    ws.is_repeat_session,
    COUNT(DISTINCT ws.website_session_id) AS sessions,
    COUNT(DISTINCT o.order_id) AS orders,
    ROUND(
        COUNT(DISTINCT o.order_id)::decimal
        / COUNT(DISTINCT ws.website_session_id) * 100,
        2) AS conversion_rate_pct
FROM website_sessions ws
LEFT JOIN orders o
    ON ws.website_session_id = o.website_session_id
GROUP BY 1;
```

17. Best Performing Campaign by Revenue per Session

```
SELECT
    ws.utm_source,
    ws.utm_campaign,
    COUNT(DISTINCT ws.website_session_id) AS sessions,
    SUM(o.price_usd) AS revenue,
    ROUND(
        SUM(o.price_usd)::decimal
        / COUNT(DISTINCT ws.website_session_id),
        2) AS revenue_per_session
FROM website_sessions ws
LEFT JOIN orders o
    ON ws.website_session_id = o.website_session_id
GROUP BY 1,2
ORDER BY revenue_per_session DESC;
```